Language-Enhanced Dual-Level Contrastive Learning Network for Open-Set Hyperspectral Image Classification

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***Abstract*— In recent years, language-supervised vision models have demonstrated impressive potential in learning open-world concepts. Some research has introduced this learning paradigm to the hyperspectral image (HSI) processing domain; however, there has been limited work integrating textual information into the hyperspectral open-set recognition task. To fill this gap, we leverage textual supervision information in open-set HSI classification (HSIC) and propose a language-enhanced dual-level contrastive learning network (LDCLNet). Specifically, we introduce a linguistic mode with prior knowledge as a supervised signal to enhance the metric distances between closed-set samples and provide supplementary semantic information for open-set samples. Second, a dual-level visual-language (V-L) contrastive learning (CL) approach, which can align visual and language embeddings separately at the instance level and manifold level, is proposed to establish a more accurate link between visual and language representations. Finally, a distance-refined open-set recognition method is proposed, which aims to effectively discover unknown class samples during testing by refining predictions of known and unknown classes. Extensive experiments and analysis on three public HSI datasets validate the effectiveness of LDCLNet.**

***Index Terms*— Dual-level contrastive learning (CL), hyperspectral image (HSI), language-supervised vision model, open-set classification.**

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I. Introduction

OWING to the high-resolution spectral information, hyperspectral image (HSI) holds great potential to improve many research tasks, such as ground object classification, scene understanding, mineral exploration, and mapping [1], [2], [3], [4]. Coupled spectral–spatial features enable HSI to recognize the surface materials, particularly those that have similar spectral signatures in visual cues [5], [6], [7]. The HSI classification (HSIC) task, as the core of remote sensing interpretation requirements, identifies land cover types by synergistically utilizing the spatial–spectral information of HSIs [8], [9], [10]. Early HSIC algorithms mainly focused on addressing the high dimensionality and nonlinearity of spectral features [11], [12]. They typically transformed highly unstructured spectral features into a structured feature space and then used linear methods for classification [13], [14], [15], [16], [17]. Despite their comprehensive theoretical interpretability, these methods heavily rely on handcrafted features so it is difficult to produce reliable classifiers due to overly ideal assumptions.

Driven by the abundance of training data and a large number of parameters, deep learning (DL) has provided new possibilities for the overall advancement of artificial intelligence (AI) with HSI [18], [19], [20]. In HSIC, many deep neural network (DNN)-based backbones have brought the promise of more refined and informed feature extraction and decision-making [21], [22], [23], [24]. These models, improved according to specific task requirements, have been employed to address various challenges in HSIC, thereby extending the application scopes of HSI. For instance, in light of the scarcity of labeled HSI data, some researchers have proposed classification methods based on few-shot learning (FSL) [25], [26], [27], domain adaptation (DA) [28], [29],

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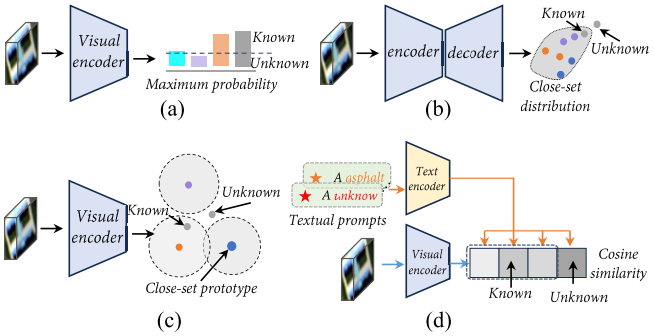


Fig. 1. Different open-set HSIC methods. (a) Probability-based method. (b) Reconstruction-based method. (c) Prototype-like method. (d) V-L-based method.

or domain generalization (DG) [30] to alleviate the burgeoning demand for high-quality labeled data.

DL-based HSIC methods have achieved remarkable performance, but they are typically trained based on a closed-set assumption, where all test classes can be seen during training. However, modern HSIC models [31], [32], [33] deployed in the open world often inevitably struggle with inputs from unknown classes, i.e., samples from categories that the network has not been exposed to during training. When encountering, such an unknown sample, traditional DNNs usually classify it as one of the known classes, leading to incorrect predictions. This will be a significant impediment to extending the model's application scenarios, which may mislead subsequent remote sensing interpretation.

To address this problem, open-set HSIC offers a viable solution to this challenge and has seen an influx of research interest. Open-set HSIC aims to train a prediction model that can accurately classify samples from known categories (i.e., closed set) while correctly rejecting samples from unknown categories (i.e., open set). A straightforward yet effective choice for open-set HSIC is to purify the unlabeled set by detecting and filtering outliers, also known as the detect-and-filter strategy. These methods typically rely on heuristic criteria and can be broadly grouped into three categories [as shown in Fig. 1(a)–(c)]: probability-based methods [34], reconstruction-based methods [35], and prototype-like methods [36], [37], [38], [39], [40]. Probability-based methods typically determine whether a sample belongs to a known or unknown class through the maximum probability of the SoftMax outputs. Inspired by extreme value theory [41], some improved methods replace the SoftMax function with the OpenMax function to fit the Weibull distribution of known classes [34]. Reconstruction-based methods follow a learning principle that unknown samples are not obeyed the observed distribution formed by the training samples, resulting in high-reconstruction errors during the testing phase. For example, to simultaneously consider the latent representations of both spatial and spectral branches, Yue et al. [35] proposed a spatial–spectral latent reconstruction framework for open-set HSIC.

Among the contenders of probability-based and reconstruction-based methods, prototype-like methods [36], [37], [38], [39] have gained widespread recognition for their capabilities. Typically, each prototype can be viewed as the mean vector of the embedded support points belonging to its class [36]. Prototype-like methods learn a metric space with high-interclass separability by directly using predefined distances, thus explicitly increasing the distance between known and unknown classes. For example, Sun et al. [42] learned a latent open space with greater distances through reciprocal points learning, hence reducing the open-space risk. Du et al. [39] employed weighted Pearson distance and feature covariance to jointly determine unknown categories. Chen et al. [43] predicted unknown classes in the fractional Fourier domain using prototype learning based on Euclidean distances.

The aforementioned open-set HSIC methods serve to enhance the model's robustness to unknown classes in open-world scenarios, albeit without improving their zero-shot inference capability. In the computer vision domain, recent vision-language models represented by contrastive language-image pretraining (CLIP) [44] exhibit remarkable capabilities in open-world perception. By leveraging language information and undergoing extensive contrastive training on large-scale text–image pairs, the CLIP model has opened up new possibilities for image classification. The goal of visual-language (V-L) models is to establish accurate relationships between visual and linguistic concepts. Compared with language signals, visual features are more natural and harder to abstract, which is a significant impediment to the transferability and generalizability of visual representations [45], [46]. By fully aligning the representations of the visual and language modalities, the inherent high compressibility and abstraction advantages of language signals will serve as additional inductive biases to guide visual tasks. Existing V-L pretrained models have already presented impressive understanding capabilities for open-world concepts [47], [48], [49], which can largely benefit from the incorporation of language models.

Inspired by this advantage, some researchers began integrating language models into HSIC tasks. A notable work is a language-aware DG network (LDGNet) [50], which for the first time introduced textual descriptions as auxiliary information into the HSIC domain. Subsequently, Li et al. [51] used textual descriptions as additional cues to guide cross-domain few-shot models in learning more robust and accurate support prototypes. Some work [52], [53] also tried to integrate text information into multisource classification tasks to assist in the collaborative classification of hyperspectral and LiDAR data. Although these studies successfully introduced language before multiple HSIC tasks, none have explored the potential benefits of text information for open-set HSIC.

To fill this gap, we introduce text information into the open-set HSIC task and develop a language-enhanced dual-level contrastive learning network (LDCLNet) for open-set HSIC. Specifically, by introducing the linguistic mode with prior knowledge as supervised information, the metric distances between closed-set samples can be improved during training. In addition, during the testing stage, the text embedding of the unknown class will also provide supplementary semantic information to assist in detecting open-set samples. To fully align visual and language embeddings, we propose a dual-level

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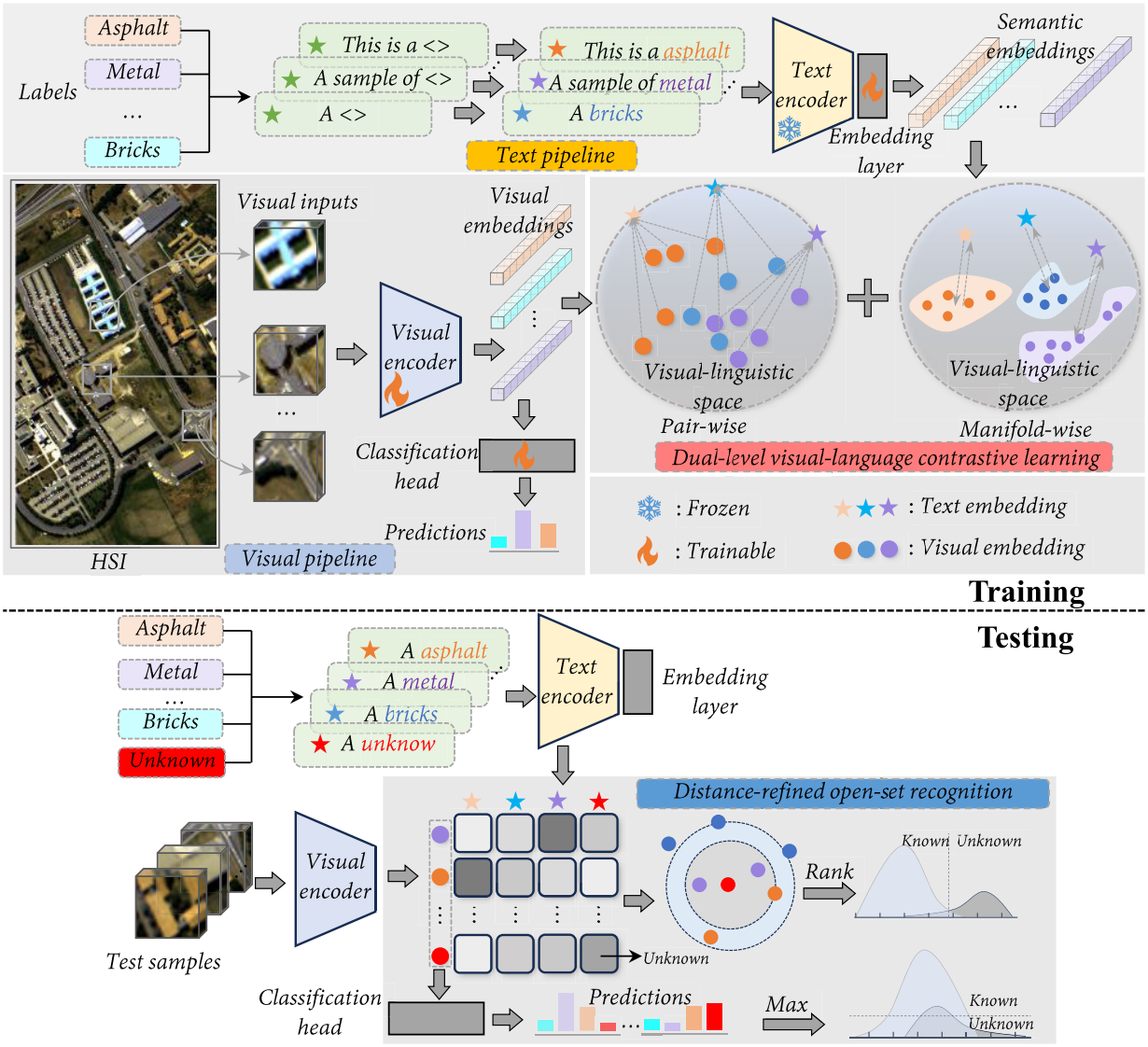


Fig. 2. Schematic of the proposed LDCLNet. The training stage mainly includes a visual pipeline to extract visual features, a text pipeline to extract semantic features, and a dual-level V-L CL module to align visual and semantic representations. In the testing stage, to classify the closed-set samples and detect open-set samples, an additional description of unknown classes is added to the text branch and a distance-refined open-set recognition method is employed.

V-L contrastive learning (CL) approach that aligns visual and language embeddings separately at the instance level and manifold level. Finally, to effectively discover unknown class samples during testing, we propose a distance-refined open-set recognition method. It refines predictions of known and unknown classes by ranking the distances between testing embeddings and well-aligned visual embeddings.

The main contributions of this article can be summarized as follows.

1) The abstract language descriptions of different samples are introduced as auxiliary prompts for open-set HSIC. Specifically, labels of known classes are converted into textual prompts to guide the training of visual features, while the text description of unknown classes is also mapped into language embeddings to aid in open-set recognition during the testing phase.
2) A dual-level vision-language CL approach is proposed, which can align text and visual embeddings at both the instance and manifold levels. By reducing the local and global vision-language representation gaps, the model will establish a more accurate link between visual features and natural language for known and unknown classes.
3) We proposed a distance-refined open-set recognition method. To balance classification outcomes between open-set and closed-set scenarios, we refine the prediction results by ranking the neighbor distances between testing visual embeddings and well-aligned visual embeddings.

The remainder of this article is organized as follows. In Section II, the proposed LDCLNet is technically presented in detail. Section III reports and analyzes experimental results. Finally, the conclusion is outlined in Section IV.

II. Methodology

The overall framework of the proposed LDCLNet is illustrated in Fig. 2. During the training phase, the labels of training samples are converted into textual prompts.

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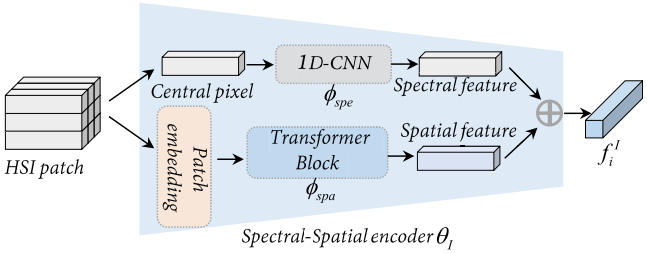


Fig. 3. Schematic of the spectral–spatial encoder.

Hyperspectral patches and their corresponding textual prompts are then input into the visual encoder and text encoder, respectively, to obtain their embeddings. To align the visual and textual representations, a dual-level CL approach is proposed to optimize the network. During the testing phase, an additional description of unknown classes is added to the text branch. Then, by calculating the visual–text similarity and using a general classifier, the test samples are classified into known and unknown classes. Finally, a distance-refined open-set recognition method is employed to purify the classification results of known and unknown samples. The detailed design of three highlighted constituent modules in the LDCLNet is presented as follows.

A. Visual and Textual Feature Extraction

In conventional HSIC methods or open-set HSIC methods, HSI is generally patched into a hyperspectral cube for network input. Following this strategy, HSI is defined as $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, width, and number of bands, respectively. After cutting the HSI, the model's input is denoted as $\boldsymbol{x}_i \in \mathbb{R}^{p\times p\times B}$, where p is the patch size. Given a training dataset $\mathcal{D}_{\text{tr}} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ containing n labeled samples and C known classes, the label of $\boldsymbol{x}_i$ is defined as $\boldsymbol{y}_i \in \mathcal{Y}_C = \{1, \ldots, C_k\}$. In a typical closed-set HSIC task, the number of categories in the test samples is consistent with the training set, i.e., there are also C classes. However, for the open-set task, assuming that the testing set with N samples is $\mathcal{D}_{\text{te}} = \{\boldsymbol{x}_{n+1}, \ldots, \boldsymbol{x}_{n+N}\}$, the number of categories in $\mathcal{D}_{\text{te}}$ will be $C + C_u$. C_u is the number of unknown classes, and during the testing phase, these will be classified as the $(C+1)$th class. For open-set recognition, the goal is to learn a model from $\mathcal{D}_{\text{tr}}$ that classifies test samples into $C+1$ classes, i.e., one of the C classes or an unknown class indexed by $C+1$.

1) Image Pipeline: As shown in Fig. 3, to obtain informative visual embeddings, we employed an existing dual-branch network [23] combining a Transformer and a 1-D convolutional neural network (1D-CNN) to extract features. The combination of 1D-CNN and Transformer can extract detailed spectral features while capturing long-range spatial dependencies [23]. Specifically, let the visual encoder be θ_I, which includes both spectral ϕ_{spe} and spatial branches ϕ_{spa}. The spatial branch consists of a patch embedding block and four Transformer blocks, where the patch size is 2 and the embedding dimension is d (set to 192 dimensions). In addition, the components of the transformer blocks are the same as those in the vanilla Vision Transformer (ViT), consisting only of multihead self-attention (with four heads), multilayer perceptron (MLP), and layer normalization (LN). For the spectral branch ϕ_{spe}, a 1D-CNN is employed to extract detailed spectral features of the central pixel. This network consists of 1-D convolutional layers, batch normalization (BN), and pooling layers. To align with the feature dimensions of the spatial branch, the output feature vector of the spectral branch also has a dimension of d. Finally, a weighted sum-based feature fusion is used to integrate the feature representations of the two branches. The formula is represented as follows:

$$\begin{aligned}\boldsymbol{f}_i^I &= \theta_I(\boldsymbol{x}_i) \\ &= \lambda\phi_{\text{spe}}(\boldsymbol{x}_i) + (1-\lambda)\phi_{\text{spa}}(\boldsymbol{x}_i)\end{aligned} \tag{1}$$

where λ is a weight parameter and $\boldsymbol{f}_i^I \in \mathbb{R}^{1\times d}$. Typically, following the setting of [23], 0.9 is chosen as the value of λ. Notably, the fusion of spatial and spectral features still follows the design of [23], where the two features are directly summed.

2) Text Pipeline: Fig. 2 illustrates the details of the text branch. Unlike HSIC algorithms based purely on visual representations, which typically convert labels into one-hot encodings for subsequent cross-entropy calculations, we use the land-cover types of labels as language supervision to aid in training the visual tasks. Specifically, following CLIP [44] and its variants [45], [46], the labels are first converted into textual descriptions using a prompt template. Taking the "a ⟨⟩" as an example, "*tree*" will be converted to "a *tree*" (the selection of specific prompt templates will be analyzed in detail in Section III-E). Subsequently, the text encoder of the CLIP model is used to encode these descriptions. Thanks to the pretraining on large-scale text–image pairs, the text encoder of CLIP has powerful text encoding capabilities. To preserve this powerful semantic understanding capability, the parameters of the text encoder are frozen. Then, to better achieve visual–text alignment for hyperspectral data, a tunable linear layer is used to project and optimize the text features. To maintain the same dimensionality as the visual branch features, the text features are also projected into a d-dimensional feature space. Mathematically, let the closed-set textual description be $\boldsymbol{y}_i$ be $t_i \in \mathcal{T}_C = \{$"*a class 1*," $\ldots$, "*a class C*"$\}$, the text encoder be ϕ_t, and the linear projection layer be ρ_p; then, the text embedding can be obtained as follows:

$$\boldsymbol{f}_i^T = \theta_T(t_i) = \phi_t(\rho_p(t_i)). \tag{2}$$

Notably, the core idea behind the CLIP model is to pretrain a multimodal model on large-scale visual–text paired data, but we just use the CLIP's language model as an additional supervisory encoder like most methods [50], [51]. In addition, our approach primarily utilizes the text encoder of CLIP, while the visual encoder is custom-designed based on the unique characteristics of hyperspectral data.

B. Dual-Level V-L CL

The technical core of these V-L models is the alignment of image–text features. Only accurate V-L relationships can take advantage of language models in visual tasks. A simple yet universal alignment method is CL [44], which directly maximizes the cosine similarity between visual embeddings and text embeddings. The pairwise V-L CL of CLIP effectively

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reduces the difference between each visual sample and its textual description. This is because each visual sample in their pretraining data is paired with a distinct language description. However, when the relationship between text and vision is nonbijection (i.e., a single-textual prototype may correspond to multiple divergent visual prototypes), optimization conflicts between single text embedding and multiple visual embeddings will arise. Especially when noisy samples are present in the visual data, they are treated as hard samples and attract more optimization energy. To address this issue, we propose a dual-level CL approach to align visual and textual embeddings at both the instance level and the manifold level. This enables textual descriptions to establish relationships with visual features from both local instances and the global manifold (distribution).

Given a pair of visual–text embeddings $\{\boldsymbol{f}_i^I, \boldsymbol{f}_i^T\}$, an image classifier for textual label set $\mathcal{T}_C$ can be implemented by computing the posterior probabilities of classes with

$$p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_C) = \frac{\exp\left(\cos\left(\boldsymbol{f}_i^T, \boldsymbol{f}_i^I\right)/\tau\right)}{\sum_{t_j \in \mathcal{T}_C} \exp\left(\cos\left(\boldsymbol{f}_j^T, \boldsymbol{f}_i^I\right)/\tau\right)} \tag{3}$$

where $p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_C)$ is the probability of textual label (or class label) t_i given HSI input $\boldsymbol{x}_i$, $\cos(\cdot,\cdot)$ is the cosine similarity metric and τ is a temperature hyperparameter. Notably, following the CL framework of the original CLIP model [44], the temperature parameter is adopted to control the logit scale in the softmax operation. Equivalently, $t_j \in \mathcal{T}_C \iff y_j \in \mathcal{Y}_C$, hence $p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_C) \iff p(y_i \mid \boldsymbol{x}_i; \mathcal{Y}_C)$. Next, for the training dataset $\mathcal{D}_{\text{tr}}$, the objective function of instance-level CL for visual can be achieved by optimizing cross entropy

$$\mathcal{L}_{\text{In}}^V = \frac{1}{n}\sum_{i=1}^{n}\sum_{t_j \in \mathcal{T}_C} -\mathbb{1}\left(t_j = t_i\right)\log p\left(t_j \mid \boldsymbol{x}_i; \mathcal{T}_C\right) \tag{4}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $\mathbb{1}(\cdot) \in \{0, 1\}$. Similarly, the corresponding loss function for texts is

$$\mathcal{L}_{\text{In}}^T = \frac{1}{n}\sum_{i=1}^{n}\sum_{\boldsymbol{x}_j \in \mathcal{D}_{\text{tr}}} -\mathbb{1}\left(\boldsymbol{x}_j = \boldsymbol{x}_i\right)\log p(\boldsymbol{x}_i \mid t_i; \mathcal{D}_{\text{tr}}) \tag{5}$$

where the computation of $p(\boldsymbol{x}_i \mid t_i; \mathcal{D}_{\text{tr}})$ s analogous to $p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_C)$, with the distinction that the visual positive–negative sample CL is replaced by its textual counterpart. Following CLIP [44], the instance-level objective function can be written as follows:

$$\mathcal{L}_{\text{In}} = \frac{\mathcal{L}_{\text{In}}^V + \mathcal{L}_{\text{In}}^T}{2}. \tag{6}$$

To reduce the distance between category clusters and their corresponding textual descriptions, a manifold-level CL is proposed. General manifold learning estimates the data manifold for each category and minimizes the point-to-manifold distance between target embeddings and category manifolds. To mitigate errors from manifold estimation, we adopt a more direct approach by minimizing the distance between textual embeddings and category means to achieve manifold-level alignment, i.e.,

$$\mathcal{L}_{\text{Mf}}^V = -\frac{1}{C}\sum_{c=0}^{C}\frac{1}{|G(i)|} \cdot \left(\sum_{i \in G(i)} \log \frac{\exp\left(\cos\left(\boldsymbol{f}_i^T, \boldsymbol{P}_i^I\right)/\tau\right)}{\sum_{j=1}^{C}\exp\left(\cos\left(\boldsymbol{f}_i^T, \boldsymbol{P}_j^I\right)/\tau\right)}\right) \tag{7}$$

where $\boldsymbol{P}_i^I$ is the visually mean vector of the ith class and $G(i) = \{i \mid \boldsymbol{P}_i^I = \boldsymbol{P}_j^I\}$. The calculation method for $\mathcal{L}_{\text{Mf}}^T$ is similar to $\mathcal{L}_{\text{In}}^T$. $\mathcal{L}_{\text{Mf}}$ is also represented as follows:

$$\mathcal{L}_{\text{Mf}} = \frac{\mathcal{L}_{\text{Mf}}^V + \mathcal{L}_{\text{Mf}}^T}{2}. \tag{8}$$

To ensure the task relevance of the visual embeddings, we also fed the visual embeddings into the classification head (including SoftMax layer) to compute the classification loss

$$\mathcal{L}_{\text{Cls}} = \frac{1}{n}\sum_{i}\mathcal{L}_{\text{ce}}\left(\rho_c\left(\boldsymbol{f}_i^I\right), \mathbf{y}_i\right) \tag{9}$$

where $\mathcal{L}_{\text{ce}}$ represents the cross-entropy loss and $\rho_c(\cdot)$ is the classification head. Finally, the overall loss can be summarized as follows:

$$\mathcal{L} = \omega(\mathcal{L}_{\text{In}} + \mathcal{L}_{\text{Mf}}) + \mathcal{L}_{\text{Cls}} \tag{10}$$

where $\mathcal{L}_{\text{In}}$ represents the instance-level CL loss, $\mathcal{L}_{\text{Mf}}$ represents the class-level CL loss, $\mathcal{L}_{\text{Cls}}$ is the classification cross-entropy loss, and ω represents the optimized weights of the visual–text alignment task.

C. Distance-Refined Open-Set Recognition

Existing open-set HSIC methods typically obtain predicted probabilities of all test samples using a closed-set classification model, and then use a threshold to decide whether a sample belongs to an unknown class or a known class. A significant advantage of V-L models for practical applications is their support for open-set classification. Since the classifier of (3) can be implemented with any class name t_i, it is possible to classify virtually any class.

To leverage this advantage, we further add a textual embedding for an unknown class during the open-set testing phase. For simplicity, we use "a *unknown*" as the textual description for the unknown class. It is important to note that our model has not been pretrained on a large scale of HSI–text pairs, making it challenging to achieve open-dictionary zero-shot inference. Therefore, we use the language model solely for detecting unknown classes. Typically, the textual descriptions set of the open-set classification task is defined as $\mathcal{T}_O = \{$"a *class 1*," ..., "a *class C*," "a *unknown*"$\}$. Then, the class of a test sample $\boldsymbol{x}_i$ can be obtained by

$$\hat{y}_i = \arg\max_{t_i \in \mathcal{T}_O} p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_O) \tag{11}$$

where the calculation of $p(t_i \mid \boldsymbol{x}_i, \mathcal{T}_O)$ is similar to $p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_C)$ in (3), i.e.,

$$p(t_i \mid \boldsymbol{x}_i; \mathcal{T}_O) = \frac{\exp\left(\cos\left(\boldsymbol{f}_i^T, \boldsymbol{f}_i^I\right)/\tau\right)}{\sum_{t_j \in \mathcal{T}_O}\exp\left(\cos\left(\boldsymbol{f}_j^T, \boldsymbol{f}_i^I\right)/\tau\right)}. \tag{12}$$

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To detect open-set samples from multiple perspectives, the output probabilities of the classification head are retained. Although this rudimentary open-set detection strategy inevitably relies on some heuristic criteria, the prediction entropy of the classifier, indeed, contains information about open-set samples to a certain extent. In LDCLNet, this heuristic design is also retained. However, we empirically set a fixed threshold (i.e., 0.2) to identify those open-set samples that are simple and easy to distinguish. Mathematically, the decision can be made via level-set estimation

$$\hat{y}_i = \begin{cases} \arg\max\limits_{y_i \in \mathcal{Y}_C} p\left(y_i \mid \rho_c\left(\boldsymbol{f}_i^I\right); \mathcal{Y}_C\right), & S_i \geq \alpha_c \\ C+1, & S_i < \alpha_c \end{cases} \tag{13}$$

where S_i is the maximum value in the prediction probability $\rho_c(\boldsymbol{f}_i^I)$.

Ideally, the above open-set detection method can identify most of the distinctly separable open-set samples. However, the open-set recognition performance of these methods typically relies on heuristic designs and is prone to sacrificing the classification performance of closed-set samples. To achieve a more refined classification result, a nonparametric nearest-neighbor distance [54] is used to refine the open-set detection results. It is noteworthy that the nearest-neighbor distance method is only adopted to refine the open-set recognition outcomes. In other words, it is merely a subcomponent of the open-set recognition strategy we propose. Denote the visual embedding set of training data as $\mathcal{F} = \{\boldsymbol{f}_1^V, \boldsymbol{f}_i^V, \ldots, \boldsymbol{f}_n^V\}$. The Euclidean distances with respect to test embedding $\boldsymbol{f}_j^{V,*}$ can be calculated by

$$d_{ij} = \left\| \boldsymbol{f}_i^V - \boldsymbol{f}_j^{V,*} \right\|_2. \tag{14}$$

Then, the distance sequence $d_{:j}$ is reordered as $\mathbb{D}_j = \{d_{(1)}, \ldots, d_{(n)}\}$. Finally, the decision function is given by

$$S\left(\boldsymbol{f}_j^{V,*}; k\right) = \mathbb{1}\left\{-d_{(k)} \geq \alpha_d\right\} \tag{15}$$

where $d_{(k)}$ is the distance to the kth nearest neighbor. The threshold α_d is typically chosen as 95% so that a high fraction of closed-set data is correctly classified. The threshold does not depend on the open-set data.

III. Experimental Results and Analysis

This section primarily presents the open-set HSIC performance of our method compared to other methods on three datasets. In addition, ablation studies and parameter analyses are conducted to comprehensively and objectively evaluate our method.

A. Datasets Description

Three hyperspectral benchmark datasets, i.e., Pavia University (PaviaU), YanCheng Wetlands (YC), and Yellow River Estuary (YRE), were used in the experiments because they have been widely employed by closed-set HSIC methods for experimental evaluation. It is important to note that these datasets have separate training and testing sets rather than randomly sampling according to a fixed percentage.

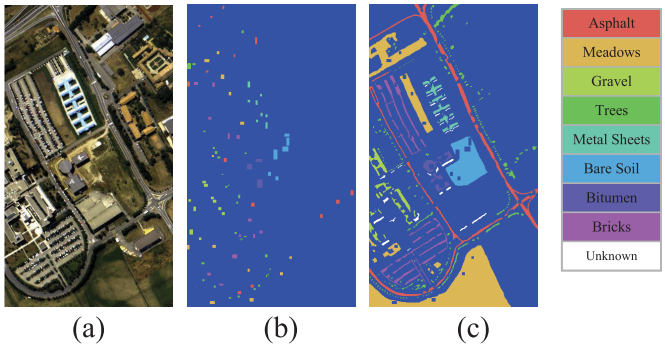


Fig. 4. Overview of the PaviaU dataset. (a) False color image. (b) Training set. (c) Testing set.

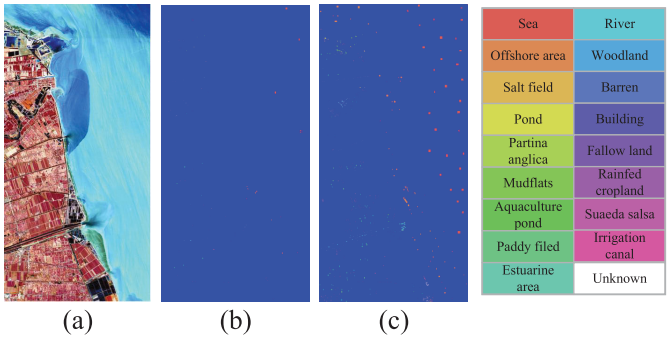


Fig. 5. Overview of the YC dataset. (a) False color image. (b) Training set. (c) Testing set.

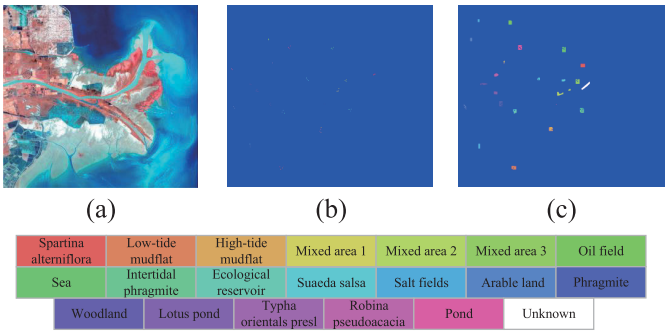


Fig. 6. Overview of the YRE dataset. (a) False color image. (b) Training set. (c) Testing set.

This is mainly because random uniform sampling will lead to potential overlap between the training and testing data, an issue highlighted in [9], [55], [56], and [57]. To avoid noncomprehensive evaluations due to model bias, we designate the last class as the open-set samples in all three datasets.

1) PaviaU Dataset: This dataset[1] covers the surrounding urban area of Pavia city and was collected by the Reflective Optics System Imaging Spectrometer (ROSIS), which has 1.3-m spatial resolution and the wavelength scope of 0.43–0.86 μm. The spatial size of the PaviaU dataset is 610 × 340 pixels, and it has 103 bands after pruning the noisy channels. The class-specific partition of training and testing sets for PaviaU datasets is reported in Table I. The false-color image and distributions of disjoint training and testing samples are shown in Fig. 4.

2) YC Wetlands Dataset: As shown in Fig. 5, this dataset includes 1175 × 585 pixels with a spatial resolution of

[1] http://dase.grss-ieee.org/index.php

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TABLE I
LAND COVER CLASSES AND NUMBER OF SAMPLES FOR THE PAVIAU DATASET

No.	Land cover type	Training Set	Testing Set
1	Asphalt	548	6304
2	Meadows	540	18146
3	Gravel	392	1815
4	Trees	524	2912
5	Metal Sheets	265	1113
6	Bare Soil	532	4572
7	Bitumen	375	981
8	Bricks	514	3364
9	Shadow (Unknown)	0	947
Total	-	3690	40154

TABLE II
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE YC DATASET

No.	Land cover type	Training Set	Testing Set
1	Sea	209	2186
2	Offshore area	140	1448
3	Salt field	6	104
4	Pond	20	173
5	Partina anglica	7	76
6	Mudflats	25	243
7	Aquaculture pond	25	238
8	Paddy filed	87	745
9	Estuarine area	27	248
10	River	27	272
11	Woodland	19	196
12	Barren	25	129
13	Building	37	489
14	Fallow land	26	208
15	Rainfed cropland	28	176
16	Suaeda salsa	14	71
17	Irrigation canal	10	28
18	Phragmites (Unknown)	0	132
Total	-	732	7162

30 m and mainly covers the coastal area of Yancheng, Jiangsu, China. YC dataset [62] was captured by the ZY1-02D AHSI sensor on April 4, 2019, and it is rich in wetland resources, with a large area of radiating sandbars, coastal forest farms, and grasslands. After removing bad bands, it contains 147 bands from the spectral range of 0.4–2.5 μm. The detailed split information of training and testing samples is summarized in Table II.

3) YRE Dataset: This dataset [62] was collected by GF-5 satellite over the Yellow River Delta in November 2018. It comprises 1185 $\times$ 1342 pixels with a spatial resolution of 30 m and 20 typical wetland classes. After moving the bad bands, there are 285 spectral bands with 150 bands ranging from 0.4 to 1.0 μm and 180 bands ranging from 1.0 to 2.5 μ remaining in the YRE dataset. Fig. 6 presents the false-color and ground-truth maps of the YRE dataset. Similarly, the training and testing sets of the YRE dataset are disjoint forms and reported in Table III.

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE YRE DATASET

No.	Land cover type	Training Set	Testing Set
1	Spartina alterniflora	84	743
2	Low-tide mudflat	84	709
3	High-tide mudflat	91	645
4	Mixed area 1	16	123
5	Mixed area 2	95	774
6	Mixed area 3	75	731
7	Oil field	72	612
8	Sea	135	756
9	Intertidal phragmite	86	758
10	Ecological reservoir	75	563
11	Suaeda salsa	67	610
12	Salt fields	81	663
13	Arable land	68	681
14	Phragmite	83	756
15	Woodland	49	502
16	Lotus pond	75	461
17	Typha orientals presl	6	30
18	Robina pseudoacacia	65	516
19	Pond	86	763
20	Yellow river (Unknown)	0	859
Total	-	1393	12255

The three datasets have distinct characteristics: 1) the PaviaU dataset has fewer classes but more labeled samples, with a relatively large number of open-set samples; 2) the YC dataset has a higher number of classes but fewer open-set samples; and 3) the YRE dataset also has a high number of classes, but it contains many open-set samples as well. Evaluating different classification methods on these three datasets with different characteristics allows for a more comprehensive assessment of their performance.

B. Experimental Implementation and Evaluation Metrics

The LDCLNet is comprised of two subnetworks: a visual encoder and a text encoder. The encoders of the textual branch employed the text encoder of CLIP, which was initialized with CLIP's pretrained weights and kept frozen during training. For the visual encoder, the spatial branch was implemented as four standard ViT layers with four heads, and the tradeoff parameter in (1) and patch size of Transformer were set to 0.1 and 2, respectively. The embedding dimensions were set to be equal to the dimension of token embedding, i.e., 512, and the features were normalized to a unit sphere, as proposed in CLIP. All experiments were implemented on the same hardware platform with GTX-3090 GPU, Intel 4210R CPU, and 45G memory. The AdamW with an initial learning rate of $1e^{-3}$ was adopted as the optimizer, and the weight decay was set to $1e^{-4}$ for the stable training of the network. Since LDCLNet is an open-set detection method, the model cannot access open-set samples during the training phase. The testing set will contain varying numbers of open-set samples. It is important to note that unknown classes within each dataset are not explicitly defined. Instead, following the experimental setup established in prior work [39], [42], we designated the

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TABLE IV
CLASSIFICATION ACCURACIES OF METHODS ON THE PAVIAU DATASET

Class	SSRN [22]	DBDA [24]	SSFTT [58]	ViT [59]	SpaSIGMA [60]	MDL4OW [61]	K-OOD [54]	LDCLNet
1	95.43±1.01	91.03±1.97	82.02±3.34	89.78±1.87	91.16±1.39	85.86±2.67	90.67±1.99	83.09±1.70
2	83.28±6.77	80.69±3.22	93.63±1.01	94.91±1.84	95.14±0.87	95.53±2.72	92.28±2.34	95.49±0.53
3	90.74±1.67	81.03±5.72	65.76±3.74	76.73±3.53	75.42±6.45	62.22±4.27	83.38±3.39	87.37±2.88
4	92.87±2.20	96.05±1.14	99.63±0.25	94.35±0.96	95.99±0.26	96.72±0.12	92.75±1.33	93.92±0.40
5	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	87.28±0.63	100.00±0.00	100.00±0.00
6	99.48±0.41	99.99±0.02	86.36±8.21	93.20±1.63	66.12±2.79	89.51±1.97	90.87±2.74	93.28±3.15
7	99.55±0.54	97.00±5.44	90.20±2.37	94.54±1.83	95.01±1.97	87.83±2.42	97.94±2.38	94.41±1.67
8	98.75±0.69	98.65±0.41	98.07±0.40	95.79±0.67	96.53±0.52	95.70±0.42	96.74±0.63	96.01±0.50
Unknown	18.67±5.11	14.81±4.95	3.84±3.79	2.28±1.19	25.96±6.88	23.46±4.58	43.15±6.52	**65.43±1.09**
OA(%)	88.70±2.97	86.52±1.77	88.50±1.05	91.07±0.74	89.00±0.50	89.81±1.68	91.06±0.77	**92.24±0.28**
AA(%)	86.53±0.79	84.36±0.83	79.95±1.41	82.40±0.36	82.37±1.28	80.46±1.35	87.53±0.81	**89.89±0.28**
Kappa	85.33±3.72	82.55±2.17	84.60±1.49	88.05±0.95	85.11±0.68	86.33±2.17	88.18±0.96	**89.67±0.36**

TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE YC DATASET

Class	SSRN [22]	DBDA [24]	SSFTT [58]	ViT [59]	SpaSIGMA [60]	MDL4OW [61]	K-OOD [54]	LDCLNet
1	81.65±9.33	91.40±5.70	97.51±1.80	100.00±0.00	99.75±0.31	96.19±5.07	100.00±0.00	100.00±0.00
2	98.34±3.31	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	99.97±0.03	100.00±0.00	100.00±0.00
3	41.73±23.79	52.31±20.47	35.96±20.41	64.81±9.21	51.73±9.72	21.35±16.55	39.62±17.68	47.89±8.30
4	69.48±12.63	79.77±2.28	80.93±2.10	78.73±8.54	71.45±6.50	76.65±2.09	73.53±3.53	74.68±9.33
5	44.47±28.11	100.00±0.00	44.47±19.40	19.74±2.35	11.05±2.44	58.95±24.51	14.21±2.55	13.16±1.86
6	80.49±10.32	91.36±1.01	86.58±3.56	88.07±2.22	77.28±6.00	84.36±1.93	88.15±0.99	89.22±1.41
7	92.52±2.64	94.45±0.72	89.33±2.92	97.14±2.31	94.87±1.34	91.77±1.24	93.70±2.14	95.55±0.63
8	79.33±10.07	97.99±0.74	99.20±0.69	92.70±1.29	95.84±1.44	99.36±0.20	95.84±1.12	97.93±0.68
9	95.89±2.11	97.50±0.16	98.47±1.26	94.11±2.35	95.32±2.86	97.98±0.72	95.81±0.97	98.39±0.36
10	55.07±16.02	87.65±2.29	98.82±0.54	91.84±2.10	88.82±3.29	95.66±1.51	94.71±2.57	88.53±3.04
11	97.14±1.76	99.39±0.75	100.00±0.00	100.00±0.00	99.49±0.79	77.04±10.67	100.00±0.00	100.00±0.00
12	84.34±10.76	95.35±1.55	95.50±2.05	95.81±1.35	84.03±2.00	73.95±4.06	93.80±2.64	93.80±1.10
13	82.66±13.64	90.35±3.04	95.95±2.05	91.70±2.23	65.15±5.15	66.46±2.00	88.43±4.44	91.33±1.94
14	63.94±8.27	77.60±2.42	94.14±1.70	89.62±2.41	85.19±2.28	89.62±0.24	79.62±3.78	91.83±2.28
15	51.48±23.35	74.77±7.89	92.39±4.75	85.00±2.70	78.98±4.24	89.43±3.53	79.55±5.98	81.82±6.73
16	63.66±2.87	68.17±1.44	70.14±6.07	64.23±1.69	60.28±2.25	70.42±2.82	62.82±1.13	61.97±2.18
17	35.71±7.82	37.86±7.35	83.57±13.48	85.71±3.19	86.45±5.71	100.00±0.00	80.00±3.64	71.43±13.55
Unknown	68.33±25.09	77.42±14.57	20.61±7.06	10.30±4.35	30.00±7.05	0.00±0.00	69.24±6.51	**85.91±4.08**
OA(%)	82.01±4.06	91.95±1.41	93.73±0.53	92.97±0.33	90.27±0.71	89.50±1.44	93.17±0.50	**94.35±0.15**
AA(%)	71.46±3.62	**84.07±1.90**	82.42±2.18	80.53±0.76	76.43±1.07	77.18±1.42	80.50±1.39	82.41±0.96
Kappa	79.13±4.55	90.53±1.62	92.56±0.63	91.65±0.39	88.45±0.84	87.58±1.64	91.89±0.59	**93.29±0.17**

last class in each dataset as the pseudounknown class. The overall accuracy (OA), class-specific average accuracy (AA), and the Kappa coefficient were employed to evaluate the classification performance. For open-set performance evaluation, we chose the class-specific accuracy of the unknown class as the metric.

C. Result Comparison and Analysis

To comprehensively verify the performance of the proposed method for open-set HSIC, seven influential methods were selected for comparison. These methods can be categorized into two aspects: conventional closed-set HSIC methods and open-set classification methods. For the former group, representative HSIC methods were chosen, including two methods based on CNN backbone: spectral–spatial residual network (SSRN) [22] and dual-attention network (DBDA) [24]; two methods based on Transformer backbone: spectral–spatial feature tokenization Transformer (SSFTT) [58] and vanilla ViT [59]; an HSI foundational model: the spatial subnetwork of hyperspectral intelligence comprehension foundation model (SpaSIGMA) [60]. In terms of the open-set classification methods, a multitask DL method for an open world (MDL4OW) [61] was chosen for comparison. Since there are few methods specifically designed for the open-set HSIC task [63], [64], [65], [66] yet, a general out-of-distribution (OOD) detection method, i.e., OOD detection with deep nearest neighbor (K-OOD) [54], was also selected.

Following a general solution for the open-set problem, closed-set HSIC methods uniformly deployed an additional SoftMax function for their networks. Next, referring to the general solution, the threshold of maximal probability was set to 0.5 to identify the unknown categories (i.e., the output with the maximal probability smaller than 0.5 will be taken as the unknown category). In addition, for credibility, all methods were initiated with hyperparameters as the corresponding paper and fine-tuned to achieve the best performance. The prompt template of the text pipeline in our method was selected as "a ⟨*land-cover type*⟩." Tables IV–VI present the quantitative experimental results of all methods on the PaviaU, YC, and YRE datasets, respectively. In addition, Figs. 7–9 illustrate the qualitative classification results of all methods on three datasets.

Based on the observation of results, several valuable conclusions are obtained. For closed-set classification algorithms, it is challenging to balance the open-set detection results with the closed-set results. Some methods, such as SSFTT [58]

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TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE YRE DATASET

Class	SSRN [22]	DBDA [24]	SSFTT [58]	ViT [59]	SpaSIGMA [60]	MDL4OW [61]	K-OOD [54]	**LDCLNet**
1	52.62±13.99	100.00±0.00	73.92±23.59	59.44±13.49	37.77±5.71	27.08±5.02	79.09±18.76	100.00±0.00
2	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
3	99.94±0.12	100.00±0.00	97.24±2.49	100.00±0.00	100.00±0.00	79.23±1.22	100.00±0.00	100.00±0.00
4	40.00±48.99	100.00±0.00	100.00±0.00	100.00±0.00	99.68±0.40	100.00±0.00	100.00±0.00	100.00±0.00
5	22.15±1.22	29.61±8.76	87.18±10.78	96.95±1.96	97.24±0.72	51.96±20.10	92.46±3.69	90.23±1.98
6	82.27±4.75	88.76±3.83	85.94±7.96	93.13±2.50	87.03±2.46	87.88±1.49	100.00±0.00	99.78±0.44
7	62.29±14.45	99.38±1.01	99.31±1.37	4.05±2.03	5.52±2.91	51.21±11.57	55.62±27.84	66.34±17.15
8	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
9	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
10	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
11	41.97±0.66	42.30±0.00	55.64±19.22	72.85±22.07	67.97±9.02	44.62±4.66	43.71±1.03	61.41±8.80
12	62.56±0.36	62.75±0.00	63.29±1.09	62.75±0.00	64.86±4.00	65.16±0.60	63.89±2.29	65.16±2.51
13	100.00±0.00	100.00±0.00	94.51±5.18	100.00±0.00	100.00±0.00	98.30±3.19	100.00±0.00	100.00±0.00
14	95.11±5.62	100.00±0.00	100.00±0.00	100.00±0.00	91.67±4.17	100.00±0.00	100.00±0.00	100.00±0.00
15	52.83±18.22	83.23±13.76	94.14±2.81	93.59±1.33	97.13±1.96	99.64±0.53	93.79±7.77	96.45±2.05
16	24.04±12.08	51.02±15.31	78.95±14.63	72.36±1.41	62.91±2.73	2.00±0.16	81.48±1.52	75.27±1.25
17	0.00±0.00	100.00±0.00	100.00±0.00	98.67±2.67	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
18	44.92±27.43	58.22±30.90	35.12±36.38	89.23±16.58	93.99±6.83	64.50±25.20	100.00±0.00	98.37±3.26
19	96.65±5.62	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	96.15±1.10	100.00±0.00	100.00±0.00
Unknown	85.05±29.90	6.59±8.74	0.63±1.10	0.05±0.09	82.56±34.88	96.72±5.05	39.70±20.47	**99.65±0.70**
OA(%)	74.84±1.43	79.13±1.79	81.36±2.68	80.01±0.95	83.56±2.56	77.89±1.78	86.11±2.91	**92.71±0.81**
AA(%)	68.12±2.62	81.09±2.06	83.29±2.77	82.15±1.06	84.42±1.89	78.22±1.73	87.49±2.39	**92.63±0.79**
Kappa	73.24±1.51	77.88±1.91	80.26±2.85	78.84±1.02	82.57±2.70	76.53±1.89	85.28±3.09	**92.28±0.86**

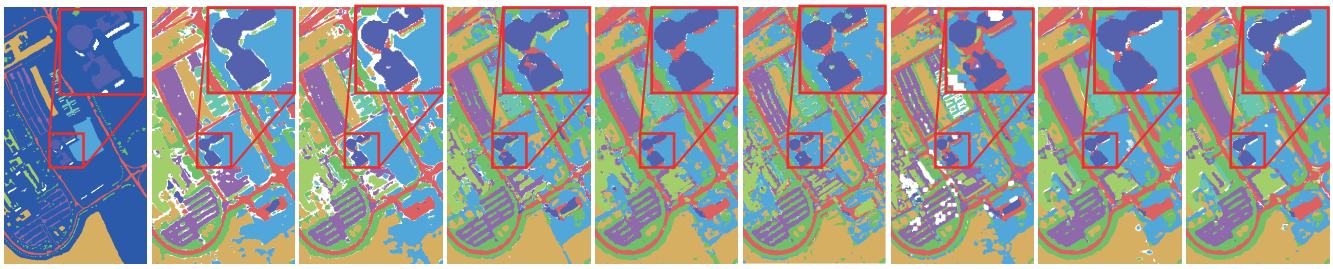


Fig. 7. Classification maps of different methods on the PaviaU dataset. (a) Ground-truth map. (b) SSRN [22]. (c) DBDA [24]. (d) SSFTT [58]. (e) ViT [59]. (f) SpaSIGMA [60]. (g) MDL4OW [61]. (h) K-OOD [54]. (i) LDCLNet.

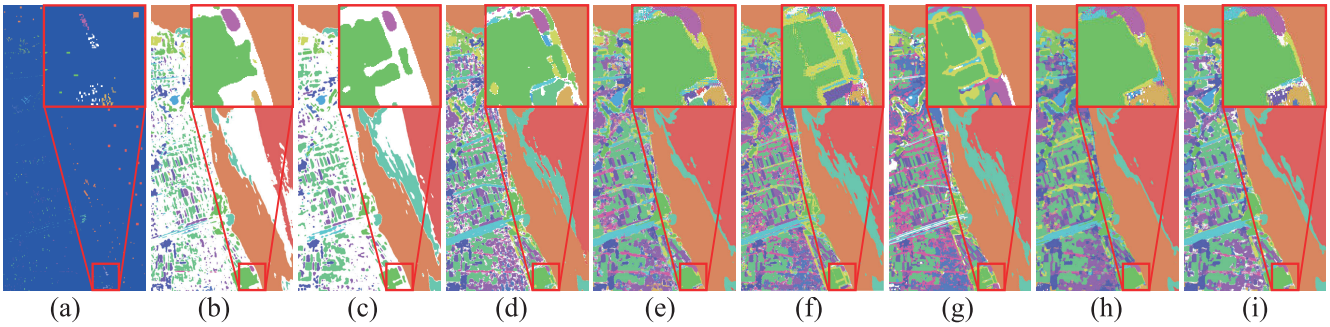


Fig. 8. Classification maps of different methods on the PaviaU dataset. (a) Ground-truth map. (b) SSRN [22]. (c) DBDA [24]. (d) SSFTT [58]. (e) ViT [59]. (f) SpaSIGMA [60]. (g) MDL4OW [61]. (h) K-OOD [54]. (i) LDCLNet.

and ViT [59], may achieve high OA but fail to detect open-set samples (i.e., the overall classification accuracy for unknown classes is relatively low). This is because they tend to predict open-set samples as known classes with high confidence, making it difficult to rely solely on a threshold for open-set detection. Open-set detection results primarily stem from two aspects: the learning of the feature space and the design of the open-set recognition strategy. Regarding the learning of the feature space, these closed-set methods are not specifically optimized for open-set tasks and depend solely on the feature extraction capability of the model. The open-set recognition strategy, on the other hand, only relies on a simple probability-decision rule.

Some closed-set methods, such as SSRN and DBDA, have shown competitive open-set detection results on the YC and YRE datasets, but this has come at the expense of sacrificing the closed-set performance. The presence of the open-set threshold makes some correctly classified closed-set samples be incorrectly predicted as open-set samples. Moreover, the open-set performance of these methods is highly unstable, exhibiting large variance in open-set accuracy and failing to identify open-set samples effectively on certain datasets.

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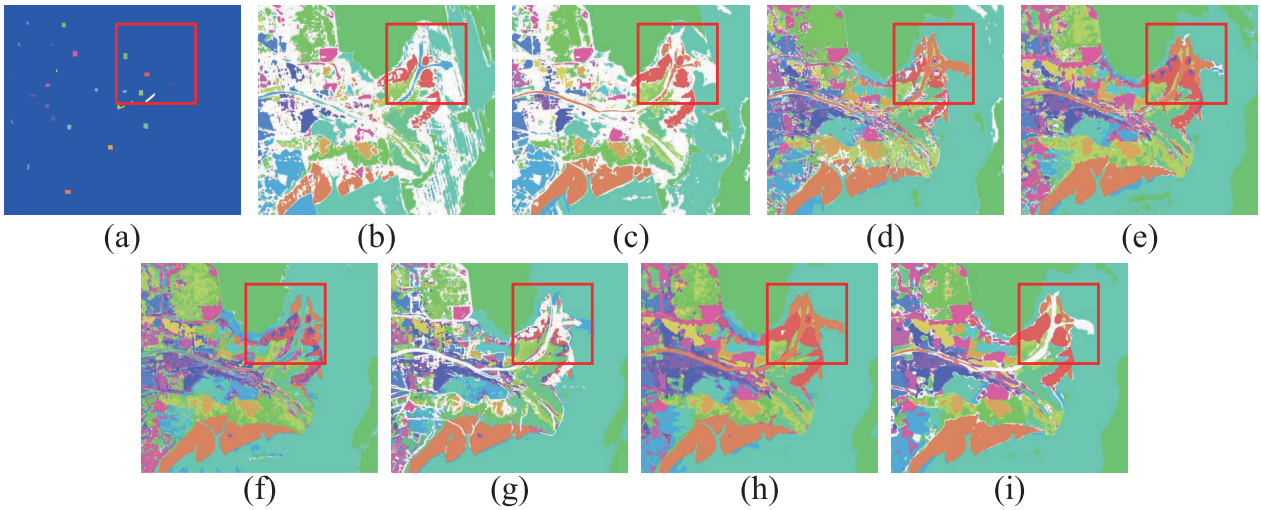


Fig. 9. Classification maps of different methods on the PaviaU dataset. (a) Ground-truth map. (b) SSRN [22]. (c) DBDA [24]. (d) SSFTT [58]. (e) ViT [59]. (f) SpaSIGMA [60]. (g) MDL4OW [61]. (h) K-OOD [54]. (i) LDCLNet.

This occurs because they did not achieve good closed-set training on those datasets, resulting in relatively low closed-set prediction probabilities. Since the 0.5 threshold is higher than the closed-set prediction probabilities, the model sacrifices a significant portion of its closed-set classification performance.

As the number of classes increases, the open-set recognition task becomes more challenging. Notably, on the YC and YRE datasets, which have a larger number of classes, the difficulty for closed-set methods to balance open-set and closed-set performance becomes more apparent. These methods struggle to achieve both high OA and unknown class accuracy simultaneously. In addition, some open-set methods may also fail when the number of classes increases. For example, on the YRE dataset, these two open-set methods do not achieve satisfactory results.

The number (proportion) of open-set samples is a determining factor in the difficulty of the open-set recognition task. When the number of open-set samples is small, i.e., with the YC dataset, the models struggle to detect these samples. A small number of open-set samples may be overshadowed by a large number of closed-set samples. Conversely, when there are many open-set samples, as in the PaviaU and YRE datasets, fully detecting all open-set samples becomes challenging. It is evident that while some methods have detected partial open-set samples, they have not achieved high accuracy.

Our method achieves the highest classification accuracy on both OA and open-set detection precision metrics. Across the three datasets with varying levels of difficulties, our method consistently delivers excellent classification performance, indicating that it does not suffer from severe model bias. For open-set detection results, LDCLNet outperforms the second-best approach by 22.28%, 16.67%, and 2.93% across the three datasets, yielding its superior open-set detection performance. In terms of OA, LDCLNet exceeds the second-best method by 1.18%, 0.62%, and 6.6% on the three datasets, respectively. This suggests that our method does not sacrifice much in closed-set classification performance. Moreover, our method exhibits lower variance in both OA and open-set accuracy, highlighting its stability and robustness. However, in some specific categories, such as the 5th and 16th classes in the YC datasets, LDCLNet may underperform compared with other methods. This could be due to the model not fully fitting the distribution of those categories, leading to some implicit model biases.

From the classification maps in Figs. 7–9, it is evident that LDCLNet not only achieves clearer classification boundaries compared with other methods but also without significantly sacrificing closed-set results. Some methods, while also achieving promising open-set detection precision in quantitative results, clearly misclassify a significant number of closed-set samples as open-set (e.g., SSRN [22] and DBDA [24] on all three datasets, as well as MDL4OW [61] on the YRE dataset). Moreover, within the red rectangles, which contain both open-set and closed-set samples, LDCLNet can accurately distinguish them. This is particularly evident in the YC dataset, where open-set samples are rare (uncommon classes); yet, our method still achieves classification results nearly identical to the ground truth. These observations indicate that our proposed method can accurately detect open-set samples while maintaining high-closed-set classification performance without overfitting to a specific dataset.

D. Ablation Study

In this section, we explore the effectiveness of LDCLNet from three aspects: text encoder, dual-level V-L CL, and distance-refined open-set recognition (OSR). A summary of the classification results for different variants across the three datasets is presented in Table VIII. It is important to note that the baseline refers to the visual encoder in LDCLNet with open-set recognition performed using a threshold of 0.5. Since LDCLNet is an open-set recognition method, we first included the distance-refined OSR module to provide a more objective evaluation of the improvements brought by the text encoder. Furthermore, to comprehensively assess the specific contributions of different modules, we designed variants that combine any two modules. However, since the dual-level V-L CL module is based on the text encoder, we were unable to

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TABLE VII
ABLATION STUDY ON THREE DATASETS

Variants	Modules			Datasets		
	Textual descriptors	Dual-level V-L CL	Distance-refined OSR	PaviaU	YC	YRE
Accuracy of unknown class	×	×	×	29.91±4.87	**92.73±3.67**	**100.00±0.00**
	×	×	✓	43.15±6.52	69.24±6.51	39.70±20.47
	✓	✓	×	7.60±0.59	13.79±7.48	71.08±17.10
	✓	×	✓	46.86±4.38	71.85±12.42	74.65±20.80
	✓	✓	✓	**65.43±1.09**	85.91±4.08	99.65±0.70
OA	×	×	×	90.18±1.31	89.78±1.14	82.98±0.66
	×	×	✓	91.06±0.77	93.17±0.50	86.11±2.91
	✓	✓	×	92.21±0.76	94.14±0.41	90.32±3.37
	✓	×	✓	91.50±0.98	93.28±0.52	90.75±1.08
	✓	✓	✓	**92.24±0.28**	**94.35±0.15**	**92.71±0.81**

TABLE VIII
ACCURACIES (%)OF THREE METHODS ON THE PAVIAU DATASET WITH VARYING OPEN-SET CATEGORIES

	Methods	Unknown class								
		No.1	No.2	No.3	No.4	No.5	No.6	No.7	No.8	No.9
Accuracy of unknown class	ViT [59]	2.46	6.71	2.51	22.91	9.51	0.94	0.71	18.06	2.28
	K-OOD [54]	47.70	10.51	4.92	27.70	46.82	11.43	1.10	16.17	43.15
	LDCLNet	**62.53**	**74.94**	**16.15**	**57.80**	**63.57**	**17.36**	**15.56**	**29.98**	**65.43**
OA	ViT [59]	78.46	52.45	86.91	81.93	90.79	83.22	86.37	**85.50**	91.07
	K-OOD [54]	83.91	54.57	86.80	**88.57**	89.51	81.06	87.04	83.43	91.06
	LDCLNet	**84.98**	**75.78**	**88.02**	85.88	**91.19**	**83.88**	**87.83**	84.19	**92.24**

obtain the variant of dual-level V-L CL with distance-refined OSR.

From the results in Table VIII, five findings can be drawn. First, LDCLNet with all modules achieves the highest OA and also attains the best or second-best open-set accuracy. Second, the baseline model achieves the highest open-set precision on the YC and YRE datasets. This is due to the large probability threshold, which sacrifices too much closed-set performance. It forces most closed-set samples to be classified as open-set class, leading to higher recognition results for open-set samples. After adding distance-refined OSR, although closed-set performance declines, OA shows significant improvements. Third, excluding the extreme open-set results of the baseline, each added component leads to varying degrees of improvement in both OA and open-set precision. This yields the effectiveness of the proposed components. Fourth, the combination of the text descriptors and dual-level V-L CL significantly enhances closed-set classification performance. However, due to the absence of an open-set recognition model, its open-set detection results are relatively worse compared with the other variants. Finally, LDCLNet not only achieves the best OA but also maintains open-set results that are not significantly worse than the most extreme baseline (i.e., achieving $+35.52\%$, -6.82%, and -0.35% improvements in unknown class precision, but surpassing baseline by $+2.06\%$, $+4.57\%$, and $+9.73\%$ in OA).

To comprehensively evaluate the open-set recognition performance of LDCLNet, an experiment was designed where each class in the PaviaU dataset was iteratively selected as the open-set class. To facilitate comparison of LDCLNet's performance, a closed-set method ViT and an open-set method K-OOD [54] were chosen as counterparts. In this experiment, each class of the PaviaU dataset was successively selected as the open-set class, while the remaining classes were treated as closed-set classes. Table VIII presents the experimental results of the three methods under different open-set class conditions. It is evident that LDCLNet yields excellent open-set recognition ability, achieving the highest open-set recognition accuracy across all classes. Furthermore, in terms of overall classification performance, LDCLNet also achieves nearly the highest classification accuracy. Although LDCLNet did not achieve the highest OA in two specific classes (fourth and eighth classes), the performance gap compared with the top method is minimal, indicating that LDCLNet maintains strong closed-set classification performance while achieving competitive open-set recognition results.

TABLE IX
DIFFERENT PROMPT TEMPLATES

	Prompt templates
T_1	None. (Baseline)
T_2	$<$ *land-cover type* $>$.
T_3	A $<$ *land-cover type* $>$.
T_4	A sample of $<$ *land-cover type* $>$.
T_5	This is a $<$ *land-cover type* $>$.

E. Impact of Prompt Templates and Parameter Analysis

To further investigate the impact of textual prompts on hyperspectral open-set tasks, various prompt templates were designed to explore their effects on LDCLNet's performance. Specifically, five different prompt templates (as summarized in Table IX) were selected for comparison. The first template,

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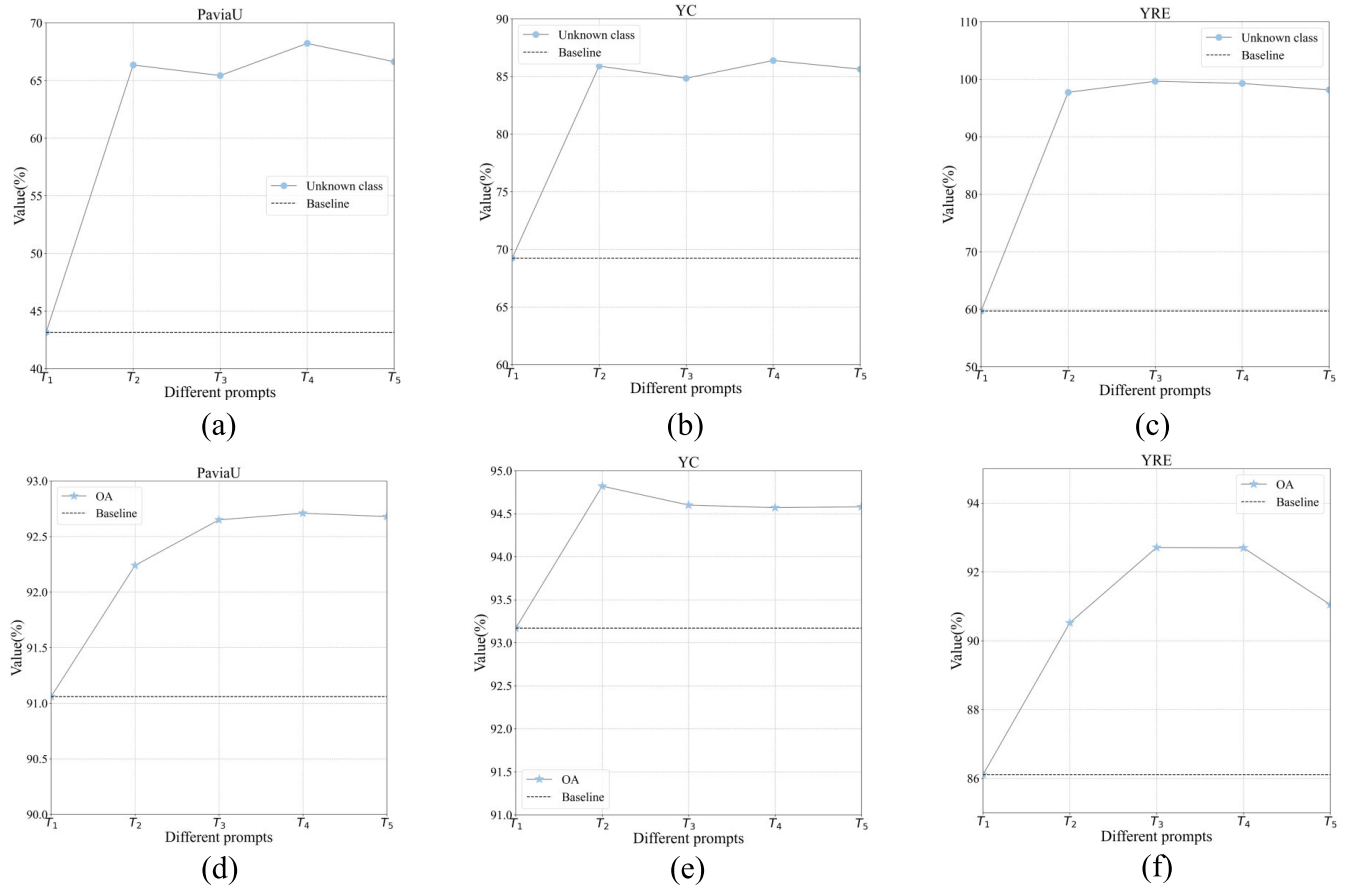


Fig. 10. Results of LDCLNet with different prompt templates in Table IX. The first row is the results of unknown class, and the second row is the results of OA. Accuracy of unknown class on (a) PaviaU dataset, (b) YC dataset, and (c) YRE dataset. OA on (d) PaviaU dataset, (e) YC dataset, and (f) YRE dataset.

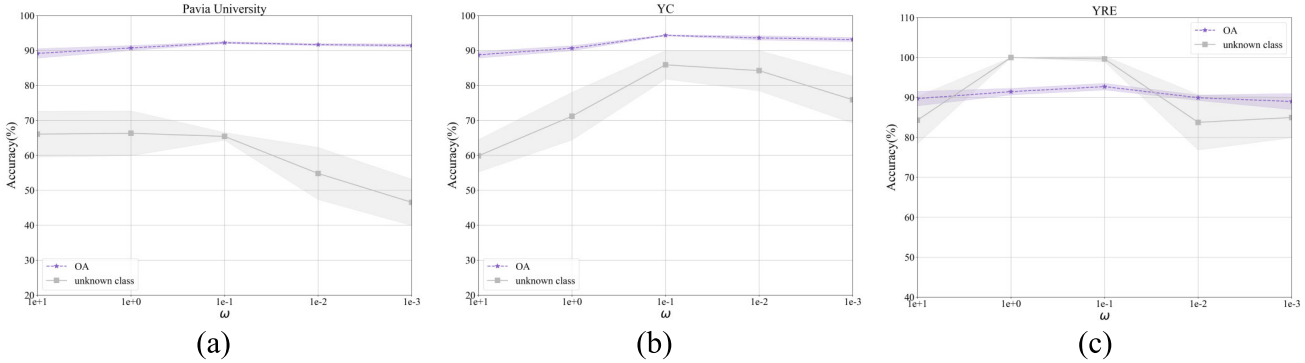


Fig. 11. Results of LDCLNet with a different ω on three datasets. (a) PaviaU dataset. (b) YC dataset. (c) YRE dataset.

"none" represents the absence of any textual prompt and can be considered the baseline model. It is important to note that, unlike previous text-enhanced HSIC methods [50], we did not add too long additional textual descriptions for training. Since the input to the visual encoder is a hyperspectral patch without surrounding contextual information, we only selected simple prompt templates with varying text lengths.

Fig. 10 illustrates the classification performance of LDCLNet with different prompt templates. The first row presents the open-set results, while the second row shows the overall performance. Adding textual prompts (i.e., the text encoder) significantly enhances both the open-set performance and OA of the model. This improvement can be attributed to the additional supervision provided by the textual information, which can offer more prior knowledge for both open-set and closed-set classification tasks. Different prompt templates have varying degrees of impact on the model's performance, but the effects on both OA and open-set performance are not significant (open-set performance varies by $\pm 2\%$ and OA varies by $\pm 1\%$). Moreover, we did not find a correlation between the model's performance and the length of the prompt text. This may be due to the limitations of the training data, as our method was not trained by large-scale image–text pairs like CLIP, which allows for a comprehensive understanding of the relationships between various visual and semantic features. Overall, the introduction of textual information has, indeed, enhanced the model's perceptual capabilities in open-set settings.

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To further explore the sensitivity of LDCLNet to the parameter w in (10), we designed a parameter experiment. Specifically, other parameters were kept constant, and the performance of LDCLNet was observed by varying the value of w within the candidate range $1e^{+1}$, $1e^{+0}$, $1e^{-1}$, $1e^{-2}$, $1e^{-3}$. Fig. 11 presents the performance of LDCLNet across three datasets under different settings of w, where OA and open-set recognition accuracy are selected as the evaluation metrics. It is clear that the model's overall classification performance shows relatively low sensitivity to the w parameter, with only a slight performance drop when w is too large (i.e., set to $1e^{+1}$). This is because the optimization of the vision-language alignment task dominates the classification task, leading to the underfitting of the decision boundary. In terms of open-set recognition accuracy, performance fluctuates when w is either too large or too small, as the imbalance between the optimization of the two tasks significantly affects the learning of the open space. However, when w is set between $1e^{+0}$ and $1e^{-2}$, the model generally can achieve promising overall classification and open-set recognition performance. Therefore, we typically set the parameter w to $1e^{-1}$ by default.

IV. Conclusion

In this article, we propose an LDCLNet, which leverages textual supervision information for open-set HSIC tasks. LDCLNet involves the linguistic mode with prior knowledge as a supervised signal to enhance the metric distances between closed-set samples, which can provide supplementary semantic information for open-set recognition. In addition, the dual-level V-L CL will build a more accurate link between visual and language embeddings by aligning visual and language embeddings separately at the instance level and manifold level. Finally, by ranking the distances between testing embeddings and well-aligned visual embeddings, the distance-refined open-set recognition method can refine predictions of known and unknown classes. Extensive experiments and analysis on three public HSI datasets yield the effectiveness of LDCLNet. Although our method achieves the best classification performance across the three datasets, it may perform slightly lower than other methods on certain specific categories. In future work, we will focus on designing a visual encoder that is more suitable for the LDCLNet framework.

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